SOLUTION OVERVIEW

A ready-to-build application platform

Four business capabilities — secure sign-in, an AI assistant, file handling and data visualisation — built, tested and deployed ahead of the event, so that competition day is spent on the idea rather than on the plumbing.

PREPARED FOR	DATE	STATUS	BASIS
Steering committee	30 July 2026	All designs approved	Four design specs

Executive summary

The competition brief is not disclosed until the day of the event. That single fact drives the whole approach: the scarce resource is not ideas or effort, it is hours. Any hour spent on set-up — creating accounts, wiring a database, configuring a deployment — is an hour not spent on the thing that will actually be judged.

This solution removes that overhead in advance. It is a single, working, publicly deployed application with the four capabilities most briefs tend to demand already in place and proven end to end. The data model is deliberately generic, so it can be reshaped to whatever the brief turns out to be without a rebuild.

Quality is not an afterthought here — automated testing is an explicitly graded part of the brief, so the test suite is treated as a deliverable in its own right, with enforced coverage thresholds and continuous integration running on every change.

4	**4**	**90%**	**£0**
capabilities working end to end	independent layers of automated testing	enforced coverage on core logic (80% overall)	AI spend in testing — no test calls a paid service

What the solution does

From a user's point of view, the platform is a single web application. A person signs in, lands on a dashboard, and can hold a conversation with an AI assistant, upload and manage files, and view their data as a chart. Each capability was designed, reviewed and approved separately, and each works against the live deployed system.

CAPABILITY 01

Secure sign-in

Users register and sign in with an email and password, held in our own database with modern password protection. Private pages are closed to anyone not signed in, and every piece of data belongs to exactly one user from the first day. Social sign-in is pre-configured and can be switched on quickly if the brief calls for it.

CAPABILITY 02

AI assistant

A conversational assistant powered by Claude. Replies appear word by word as they are generated, conversations are saved and listed in a sidebar, and each is given a short descriptive title automatically. Users can return to past conversations or delete them. Which AI model is used is a configuration setting, not a code change.

CAPABILITY 03

File upload

Users upload images, PDFs and CSV or text files up to 4 MB from the dashboard. Files are stored in managed cloud storage and listed with name, size, type, date, a thumbnail for images and a delete action. File type and size are checked on the server, not trusted from the browser.

CAPABILITY 04

Trends and charting

A "Trends" section on the dashboard plots two data series over time, with a 7 / 30 / 90-day range filter. A new user sees a clear empty state and a single button that loads a realistic sample data set. The two series colours were validated for colour-vision deficiency and for both light and dark backgrounds.

How it fits together

One application serves both the screens people see and the services behind them. It talks to three things: its own database, the AI provider, and cloud file storage. Keeping it to a single application means no separate systems to keep in step and no second deployment to manage — a material saving in a time-boxed build.

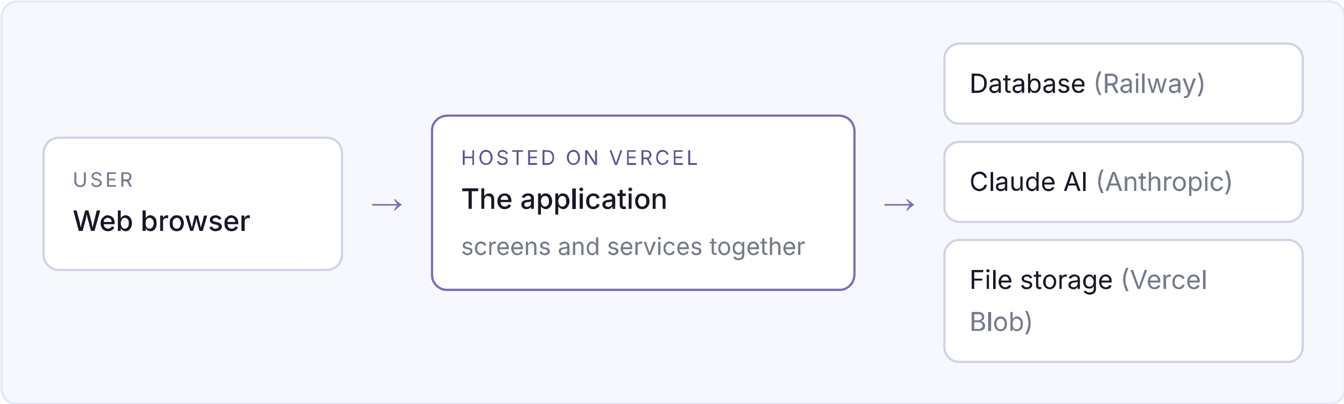

Figure 1 — The solution at a glance. Every external service is reached through a single, replaceable connection point, so swapping a supplier later is a contained change rather than a rewrite.

Delivery status

The work was sequenced so that each capability builds on a working, deployed base rather than being integrated at the end. All four designs are approved.

WORKSTREAM	DELIVERS	DESIGN APPROVED
Foundation and sign-in	Application, database, hosting, accounts, protected pages, test framework, CI	25 Jul 2026
AI assistant	Multiple saved conversations, live streamed replies, automatic titles, delete	29 Jul 2026
File upload	Upload to cloud storage, file list with thumbnails, delete, server-side validation	29 Jul 2026
Trends and charting	Two-series line chart, date-range filter, empty state with sample data	30 Jul 2026

Confidence and quality

Thorough automated testing is a graded requirement of the brief, and the approach reflects that. Four independent layers of tests check the solution from different angles, so a fault has to escape all four to reach a live demonstration.

LAYER	WHAT IT PROVES
Unit	Individual rules behave correctly — password checks, file-type limits, data calculations
Component	On-screen pieces render and respond as intended — the chat view, the upload panel, the chart controls
Integration	Services work against a real database, including the rule that one user can never see another's data
End to end	A real browser completes the whole journey: register, sign in, see the chart, upload a file, receive an AI reply

Every change is checked automatically. Continuous integration installs, checks, tests and builds the solution on every change, and publishes the coverage and browser-test reports as evidence. Coverage below the agreed threshold fails the run.

Tests cost nothing to run. No test calls the AI provider or cloud storage for real, so the suite spends no money, needs no network access, and cannot fail because a supplier was slow — the same result every time, for anyone who runs it.

Decisions worth knowing

DECISION	WHY IT MATTERS COMMERCIALLY
One application, not several	One thing to build, test and deploy — no interfaces between systems to keep aligned under time pressure
A deliberately generic data model	A labelled numeric series can become sensor readings, transactions or scores on the day without redesign
Accounts held in our own database	No dependency on a third-party identity provider, no per-user cost, no supplier account to procure
Each supplier behind one connection point	Replacing the AI provider or storage supplier is a contained change, and testing stays free of supplier costs
Deployment rehearsed in advance	Going live for the first time under time pressure is a common cause of failure; that risk is retired before the day
Existing hosting accounts reused	Both platforms are already proven on a prior project — no procurement, no new commercial relationship

Risks and how they are managed

RISK	MITIGATION
The first deployment fails on the day	The full deployment is rehearsed beforehand, with all four capabilities verified working in the live environment
The generic data model does not suit the brief	The generic table is trivially replaceable, and the tooling makes reshaping the data structure quick and low-risk
The AI provider is slow, busy or unavailable	Busy and overloaded responses are shown as a plain "try again" message rather than an error; tests never depend on the provider
The database is overwhelmed during judging	Connection limits are capped explicitly in the live configuration to prevent exhaustion under concurrent use
Quality gates slow the team down on the day	Coverage thresholds apply to the full formal run; day-of work uses a faster check and tightens afterwards
Trial environments share the live database	A deliberate, documented trade-off for a solo build: trial environments can read and write, but can never reshape the live data structure

What is deliberately excluded

Scope has been held tight on purpose. The following are known, considered omissions rather than gaps:

- Domain modelling for a specific problem — the brief is unknown, so the data model stays generic by design.
- Multi-developer working practices — this is a solo build, so shared environments and onboarding material are out of scope.
- Production concerns beyond demonstration scale — scaling, backup policy, monitoring tooling and rate limiting.
- Uploads larger than 4 MB, drag-and-drop and progress bars; renaming conversations; importing chart data from a file.
- The final visual design — a single styling pass is planned across the whole application rather than per capability.

For the committee

No decisions are outstanding: all four designs are approved and the remaining work is implementation against them. Three points are offered for awareness rather than action — the shared database between trial and live environments, the 4 MB upload limit inherent to the hosting platform, and the visual design pass still to be scheduled.

The measure of success is simple: on the day the brief is revealed, the team should be able to open a working, deployed, tested application and begin building the idea within minutes.